Feiran Wang

+1-217-305-3694 | wfr2099@gmail.com
Homepage | Google Scholar | LinkedIn
Chicago, Illinois - 60616, United States

Research Focus & Interests

- **Robotics**: Vision-Language-Action Model, World Action Model, Long-Horizon Manipulation
- **Spatial Intelligence**: Spatial Reasoning, Dynamic 3D Reconstruction, Self-Improving Agents
- **World Models**: Video Diffusion Model, Controllable Generation, Synthetic Data Generation

Education

- **University of Illinois Chicago** — *Aug. 2026 – Current*
  Ph.D. Student in Computer Science, Advisor: Prof. Yan Yan — Chicago, United States
  Prior doctoral study at Illinois Institute of Technology — *Jan. 2024 – Jul. 2026*
- **University of Michigan Ann Arbor** — *Jan. 2026 – May 2026*
  Visiting Ph.D. Student, Advisor: Prof. Dawen Cai — Ann Arbor, United States
- **University of Illinois Urbana-Champaign** — *Aug. 2022 – Dec. 2023*
  ***M.Eng. in Robotics**, Advisor: Prof. David Forsyth* — Urbana, United States
- **Shanghai University** — *Aug. 2018 – Jul. 2022*
  B.S. in Computer Science, Advisor: Prof. Xiaoqiang Li — Shanghai, China

Publications

*=Equal contribution | ‡=Author mentored by me

Robotics & Navigation

[1] **Feiran Wang**, Zezhou Shang, Gaowen Liu, Yan Yan. **RayMap3R: Inference-Time RayMap for Dynamic 3D Reconstruction**. **ECCV 2026**.

[2] **Feiran Wang**, Junyi Wu, Dawen Cai, Yuan Hong, Yan Yan. **CogniMap3D: Cognitive 3D Mapping and Rapid Retrieval**. **ICLR 2026**.

[3] Zhe Tao‡, **Feiran Wang**, Gaowen Liu, Ramana Rao Kompella, Yan Yan. **ForeAct3D: Policy-Grounded Future World Modeling for Vision-Language-Action Policies**. **Under Review, 2026**.

[4] Wenxin Rong‡, Haoxuan Wang, **Feiran Wang**, Yasutomo Kawanishi, Yan Yan. **Seeing Farther to Act Now: Long-Horizon Future Modeling for Robot Manipulation**. **Under Review, 2026**.

[5] Junyi Wu, Van Nguyen Nguyen, Benjamin Planche, Jiachen Tao, . . . , **Feiran Wang**, Terrence Chen, Yan Yan, Ziyan Wu. **Consistent Instance Field for Dynamic Scene Understanding**. **CVPR 2026**.

[6] Jing Gao‡, Wei Wang, **Feiran Wang**, Yan Yan. **LIST3R: Long-sequence Instance-aware 3D Reconstruction**. **Under Review, 2026**.

Spatial & Self-Evolving Agents

[7] **Feiran Wang**, Xiaoqi Wang, Ziwei Li, Wenbin He, Yan Yan, Liu Ren. **From Sight to Foresight: Predictive Spatial Reasoning in Vision-Language Models**. **Under Review, 2026**. *Bosch internship project*

[8] Vickie Chen‡, **Feiran Wang**, Yasutomo Kawanishi, Yan Yan. **SEVA: Self-Evolving Vision-Language-Model Agent for Video Segmentation**. **Under Review, 2026**.

3D Vision & World Modeling

[9] **Feiran Wang***, Jiachen Tao*, Junyi Wu*, Haoxuan Wang, Bin Duan, Zongxin Yang, Yan Yan. **X-Field: A Physically Informed Representation for 3D X-ray Reconstruction**. **NeurIPS 2025 (★Spotlight★)**.

[10] **Feiran Wang**, Bin Duan, Jiachen Tao, Nikhil Sharma, Gaowen Liu, Dawen Cai, Yan Yan. **ZECO: ZeroFusion Guided 3D MRI Conditional Generation**. **MVA 2025 (Oral)**.

[11] **Feiran Wang**, Bin Duan, Junyi Wu, Gaowen Liu, Yan Yan. **Kepler4D: Controllable Future Video Generation via 4D Scene State Evolution**. **Under Review, 2026**.

[12] Jiachen Tao, Benjamin Planche, . . . , **Feiran Wang**, . . . , Terrence Chen, Yan Yan, Ziyan Wu. **From Particles to Fields: Reframing Photon Mapping with Continuous Gaussian Photon Fields**. **Under Review, 2025**.

[13] **Feiran Wang**, Xiaoqiang Li, Jitao Liu. **PCCN-RE: Point Cloud Colourisation Network Based on Relevance Embedding**. *IET Computer Vision*, 2022.

Experience

- **Research Scientist Intern** *May 2026 – Aug. 2026*
  Bosch Center for Artificial Intelligence, Sunnyvale, CA, Intern
  - **Predictive Spatial Reasoning**: Developed SpatialMind, a VLM combining metric depth adaptation, geometry-motion fusion, and progressive state-chain reasoning to advance spatial intelligence.
  - **Data Engine & Benchmark**: Built a geometry-grounded data engine for driving and egocentric videos, generating **30,677 training QA pairs** and a **2,000-question benchmark** covering eight spatial reasoning tasks.
  - **Model Performance**: Achieved a **52.65%** overall score on SpatialMind-2K, **+6.95 percentage points** over the same-data Qwen3-VL-8B SFT baseline, with competitive zero-shot results on VSI-Bench, OSI-Bench, and VLM4D.
- **Robotics Research & Development Engineer** *May 2023 – Aug. 2023*
  Foxconn Interconnect Technology, San Jose, CA, Intern
  - **Simulation Development Platform**: Built Blender-to-MuJoCo toolchain for automatic URDF generation with joint dynamics calibration and collision geometry definition for rapid prototyping.
  - **Motion Planning & Control**: Integrated MoveIt for trajectory planning and collision detection in MuJoCo; designed PD controllers with real-time state feedback for precise trajectory tracking; developed visualization dashboard for monitoring joint positions, velocities, and torques.
  - **Sim-to-Real Deployment**: Optimized control parameters through simulation-hardware closed-loop iteration; validated grasping tasks on low-cost 3D-printed hardware platform.
- **Algorithm Engineer** *Apr. 2022 – Aug. 2022*
  NPLACE Startup, Shanghai, China, Intern
  - **3D Reconstruction**: Independently explored pure vision reconstruction as alternative to company's LiDAR solution; built end-to-end pipeline through MVS reconstruction to point cloud completion.
  - **MVS Network Optimization**: Applied network pruning and multi-scale feature aggregation to MVSNet; compressed model via knowledge distillation achieving 10% inference speedup.

Projects

- **Autonomous Vehicle Perception System** *Sep. 2022 – Dec. 2022*
  UIUC, *Advised by Prof. David Forsyth* Project Video
  - **End-to-End Autonomous Driving System**: Built complete ROS-based autonomous driving system on real vehicle platform covering perception, planning, and control.
  - **Multimodal Perception & Localization**: Integrated YOLO detection for obstacle recognition and emergency braking; combined learning with Kalman filtering for lane tracking; implemented LiDAR SLAM.
- **3D Medical Image Analysis** *Sep. 2024 – May 2026*
  University of Michigan Ann Arbor, Prof. Dawen Cai Project Page
  - **3D Medical Data Generation**: Designed ControlNet-conditioned diffusion model for controllable 3D volume generation, achieving **50**× synthetic data expansion to address medical image annotation scarcity.
  - **Semi-Automatic Annotation Platform**: Built interactive 3D annotation tool based on nnInteractive and nnUNet with morphological processing and connected component analysis, improving annotation efficiency.
  - **Large-Scale Distributed Training**: Deployed **240-GPU** multi-node training environment on HPC cluster with MPI optimization for large-batch parallel processing.

Skills

- **Robotics & Simulation:** ROS, MuJoCo, MoveIt, URDF, Blender, Motion Planning, PID Control
- **Programming Languages:** Python, C++, C#, MATLAB, JavaScript
- **Deep Learning & Training:** PyTorch, CUDA, TensorFlow, HPC Distributed Training
- **3D Vision & Reconstruction:** Open3D, OpenCV, SLAM, NeRF, 3DGS, VR

Academic Services & Activities

Conference Reviewer: ICLR 2026, NeurIPS 2026, ECCV 2026, CVPR 2026, ICMR 2026, ICCV 2025, MVA 2025

Journal Reviewer: Computer Vision and Image Understanding, Computers & Graphics

Invited Talk: Midwest Machine Learning Symposium (MMLS), 2025: Medical Imaging 3D Reconstruction

Honors and Awards

Cyrus Tang Scholarship, Illinois Institute of Technology 2024 – 2025
International Exchange Scholarship, Shanghai University 2020 – 2021
Academic Excellence Scholarship, Shanghai University 2019 – 2020